STAFF REVIEW · H446 081 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. Contiguous segment already in ascending order.

2. Inserts each next key into its sorted prefix.

3. Work scaling with n times log n.

4. mid3=7 then mid5=11; return index5.

I Do — source worked steps

1. Method and result — Pairs [3,8] and [1,6] → [3,8,1,6]; merge fronts → [1,3,6,8].

We Do — guided CHECK

1. Method and result
Pair sort [2,5,4,7]; merge sorted runs → [2,4,5,7].

You Do — source mark schemes

1. After width1 [2,7,1,9] [3]; after width2 [1,2,7,9] [3].

2. Compare2/1 choose1 [2], compare2/9 choose2 [1], compare7/9 choose7 [1], append9 [1], result [1,2,7,9] [1].

3. Merge O(n log n) [2], typical O(n) extra [1]; insertion $O(n^2)$ worst [2], O(1) extra [1].

Exit Ticket — source answer

1. False. It commonly uses O(n) extra storage.